

Allan Deviation Made Simple.

Dr Mark Hosey. 1st May 2019.

Introduction.

Allan Deviation is a statistical tool first used in the 1960's to study noise associated with atomic clocks and their oscillators but which has since been applied to a wide variety of applications in noise analysis.

A fairly large number of explanations and descriptions of Allan Deviation can be found on the internet along with a few academic papers that discuss the subject with reference to specific applications. In general, the authors of these publications assume their readers have a sound grasp of some basic statistics and its application to noise theory and most of them deal with the subject from a clock and oscillator (phase and frequency) perspective. But for many of those interested in understanding Allan Deviation these subjects and the associated mathematics may be long forgotten or completely unfamiliar and not in context with their particular application. As a result, a lot of these explanations and their arguments may, at first sight, appear complex and can present a challenge to fully understand.

The purpose of this essay is to present the concept of Allan deviation in simple general terms. Beginning with a short discussion of noise, it develops from first principles the statistical equations used to quantify noise and the basic expressions used in Allan Deviation noise analysis. It goes on to show how these equations can be applied to noise analysis for sensors and systems operating in the time domain, which is a familiar regime in many engineering and scientific disciplines. It then briefly discusses the analysis of frequency and phase noise in frequency dependent systems such as oscillators and clocks, this being the original application of Allan Deviation. Finally, having grasped the concepts presented, the reader is referred to several publications recommended as additional reading.

Noise Measurement.

Consider a calibrated sensor or system with transfer function 'g' which has no d.c. offset or error on its output.

With no input stimulus or signal we might naively expect its output to be zero (or null).

However the output of all measurement systems exhibit noise, the amplitude of which can often be significant in comparison to the quantity being measured. This noise contributes to the overall error in any measurement of the system's output.

Relative to any signal of interest noise can be considered to be, on some level, random and unpredictable. However in most cases, when measured regularly over a long period of time (longer than the period of the lowest frequency of interest in the noise frequency spectrum) it can be demonstrated that the noise amplitude is distributed evenly about some average.

Noise measurements made on the output of a sensor or system with no applied input will, in most circumstances, be distributed evenly about the sensor or system's null output and

calculation of the average output based on measured samples will, in most circumstances, converge toward zero as the number of samples in the average increase.

[The qualification “in most cases and circumstances” is necessary as there are troublesome noise phenomena present in the output of, for example, clocks and oscillators that are not convergent and that, regardless of the number of samples measured or the length of time over which measurements are made the mechanisms or algorithms used to determine the average of the output will not converge to a finite and well defined best estimate. Such noise phenomena are central to the discussion of Allan Variance that will be introduced and discussed a little later in this essay.]

Although the long term average noise amplitude on the output of our zero-input system may approach zero, there will nevertheless be an observable noise signal on its output with measurable power and frequency distribution.

If a perfectly stable and noise free input, denoted by v , is applied to the calibrated sensor system its output will sit at some corresponding offset from zero, determined by the sensor or system’s transfer function g , superimposed with noise that obscures the input parameters corresponding true output value:

$$V_o = v g + V_{noise}$$

or

$$V_o = V_i G + V_{noise}$$

where the input parameter is represented by a voltage equivalent input V_i which is subject to a system gain (or attenuation) G .

As before and in most cases, over some appropriate time span, the noise will be observed to be distributed evenly about the output value $V_i G$. By averaging over a long enough period of time the output noise will, in most circumstances, tend toward zero revealing the input parameters true output value which will correspond to the mean, $V_i G$. Under these circumstances the mean of the output is considered to be the best measure of the input parameter value.

The act of taking a moving (time) average of any signal that has random noise superimposed upon it will, in most circumstances, over a long period of time, reduce the noise to zero, i.e. a time longer than the period of the lowest frequency component in the noise frequency spectrum. However, noise components lying within the band of frequencies occupied by the signal of interest cannot be attenuated by determining the moving (time) average without also attenuating the same frequency components in the signal of interest. So where the input parameter is time dependent and any noise is out of band, then provided the moving average period is shorter than the period of the highest frequency in the measured parameters spectrum the output will (in the limit) be proportional to the measured input parameter.

In systems that exhibit repetition, in-band noise can be removed by averaging repeated measurements. For instance, in a repeatable experiment in which noise and the signal of interest occupy the same band of frequencies, by performing the experiment several times the results can be ‘time averaged’ by superposition of several experimental results.

In the physical sciences the standard deviation or root-mean-squared deviation is a statistical operation (calculation or algorithm) that returns a value proportional to the amount of variation in a set of measured data points or sampled values. The return of a low figure for the standard deviation indicates that the values of the measured quantity are, in general, close to the calculated mean of a set of measured values, whilst a high figure indicates that the measured values are, in general, spread over a wide range of values about the mean. In application to noise it represents a measure of the averaged noise magnitude.

It is important to note that the standard deviation does not give any indication about the convergence to the average of the data set of sampled values being operated on by an averaging algorithm or estimator.

Variance and Deviation.

The arithmetic average or mean of a discrete measurable parameter is calculated by dividing the sum of a set of its measured values by the number of measurements made (or elements in the set).

$$\bar{x} = \frac{1}{N} \sum_{n=1}^N x_n$$

Alternatively, if the parameter is a time dependent variable, say $x(t)$, and if $x(t)$ is a continuous function then the average can be calculated, in the limit as $\Delta t \rightarrow 0$, by the integral of $x(t)$ over some range of time, divided by the time span over which the average is to be determined:

$$\bar{x} = \frac{1}{\tau} \int_0^{\tau} x(t) dt$$

In general, the greater the number of samples in the set or the longer the integration time, the closer the value $\bar{x}$ will come to some stable value that can be considered the ‘true’ value of x .

$\bar{x}$ can be referred to as an average, a mean, an estimate, an “expectation” or an “expectation value” of x . When referred to as an expectation the above calculations are often represented by the mathematical shorthand:

$$\bar{x} = E[x(t)] = \langle x(t) \rangle$$

Measurements of real quantities, whether continuous or discrete, in general exhibit some distribution about a mean value, with most being clustered close to the mean or expected value. A good measure of this distribution is given by the root-mean-squared (or standard) deviation.

Where x_n is the n^{th} measurement of $x(t)$ and $\bar{x}$ is the mean, then $x_n - \bar{x}$ represents the error in the measured value x_n or, with more relevance, its instantaneous noise amplitude.

By calculating the square root of the average of the set of all deviation values squared, the standard deviation (also referred to as the Root Mean Squared deviation) can be calculated for any set of signal measurements made on the output of a sensor or system:

$$\sigma = \sqrt{\frac{\sum_{n=1}^N (x_n - \bar{x})^2}{N}}$$

Its relevance to noise is that it is a measure of the magnitude of the average amplitude of noise present on the output of a sensor or system. Since the mean of the output signal is the best measure of the signal amplitude on the sensor/system output, then the standard deviation of the signal from the mean or expected value is a measure of the amplitude of noise that is superimposed on the signal.

A much more convenient measure of noise is the mean-squared deviation as it is proportional to the *noise power*. The mean-squared deviation, also referred to as Variance, is simply given by:

$$\sigma^2 = \frac{\sum_{n=1}^N (x_n - \bar{x})^2}{N} = \frac{1}{N} \sum_{n=1}^N (x_n - \bar{x})^2$$

When N is the population number or total number of samples, this is often referred to as the population variance. However, where the number of samples N is a subset of the sample population an adjusted value of σ^2 , commonly referred to as the *sample* variance, is given by:

$$\sigma^2 = \frac{1}{N-1} \sum_{n=1}^N (x_n - \bar{x})^2$$

(The adjusted value for the sample variance accounts for the error or variance in the sample mean which will vary, depending on the specific subset of samples from the population used to calculate the sample mean. In contrast the population mean is based on all population samples and is therefore single valued and without error or variance.

Justification for the $(N - 1)$ term used to calculate the adjusted value is given in the article [Proof of Unbiasedness of Sample Variance Estimator: Proof of Unbiasedness of Sample Variance Estimator](#))

It should be noted that the variance represents an 'expectation value' as it represents an average value for the error or the expected deviation from the norm of the quantity $x(t)$.

Therefore σ^2 can be represented by the mathematical shorthand:

$$\sigma^2 = E[(x - \bar{x})^2]$$

or alternatively

$$\sigma^2 = \langle (x - \bar{x})^2 \rangle$$

Noting that $\bar{x}$ is itself an expectation of the value of x , σ^2 can be simplified:

$$\begin{aligned} \sigma^2 &= E[(x - E[x])^2] \\ &= E[x^2 - 2xE[x] + E[x]^2] \end{aligned}$$

$$= E[x^2] - 2E[x]E[x] + E[E[x]^2]$$

but the expectation value of an expectation value **is** the expectation value, i.e. the average of an average is just the average, therefore:

$$\sigma^2 = E[x^2] - 2E[x]E[x] + E[x]^2$$

$$= E[x^2] - 2E[x]^2 + E[x]^2$$

$$\sigma^2 = E[x^2] - E[x]^2$$

or alternatively

$$\sigma^2 = \langle x^2 \rangle - \langle x \rangle^2$$

This discrete measure of noise is useful but readers should note that it does not provide any information on how noise can be reduced or how the system can best be configured to optimise signal to noise ratios.

Many other algorithms have been developed for particular applications and circumstances that will calculate a figure for the deviation of a parameter or quantity of interest from its expectation value (which may be a time average, a sample average, an empirical estimate or just a best guess).

Noise reduction and its optimisation in discretely sampled systems requires characterisation of the noise **with regard to the system's sampling frequency and averaging figures**. Noise characterisation can provide knowledge of the noise frequency spectrum and may reveal individual components of noise, such as bands of frequencies present and their characteristic roll offs. Two algorithms that are well suited to noise characterisation in oscillators and clocks are N-Sample variance and 2-Sample variance.

N-Sample Variance

Where it is understood from above that for a population variance:

$$\sigma^2 = \frac{1}{N} \sum_{n=1}^N (x_n - \bar{x})^2 = E[x^2] - E[x]^2$$

Then

$$\sigma^2 = \frac{1}{N} \sum_{n=1}^N x_n^2 - \left[\frac{1}{N} \sum_{n=1}^N x_n \right]^2$$

$$\sigma^2 = \frac{1}{N} \sum_{n=1}^N x_n^2 - \left(\frac{1}{N} \right)^2 \left[\sum_{n=1}^N x_n \right]^2$$

$$\sigma^2 = \frac{1}{N} \left(\sum_{n=1}^N x_n^2 - \frac{1}{N} \left[\sum_{n=1}^N x_n \right]^2 \right)$$

For a sample variance, where we must account for inaccuracy in the computed deviation due to the low number of samples in comparison to the population number of samples:

$$\sigma^2 = \frac{1}{N-1} \left(\sum_{n=1}^N x_n^2 - \frac{1}{N} \left[\sum_{n=1}^N x_n \right]^2 \right)$$

This is the commonly used expression for N-Sample Variance.

It should be borne in mind that each sample may be measured by a process of averaging. Where the elements in the set of sample averages will be denoted by $\overline{y}_n$, then:

$$\sigma_y^2 = \frac{1}{N-1} \left(\sum_{n=1}^N \overline{y}_n^2 - \frac{1}{N} \left[\sum_{n=1}^N \overline{y}_n \right]^2 \right)$$

Where the $\overline{y}_n$ are functions of sampling time (τ) and are given by either

$$\overline{y}_n = \frac{1}{\tau} \int_{nT}^{(n+1)T} x(t) dt \text{ for a continuous function } x(t)$$

or by

$$\overline{y}_n = \frac{1}{M} \sum_{m=1}^M x_{n+m} \text{ for } N \text{ discrete sets of } M \text{ overlapping samples of } x;$$

Where $n \in [1, N]$ and $N + M$ is the total number of samples

Two-Sample Variance

Where only two samples are considered the N-Sample Variance reduces to:

$$\begin{aligned} \sigma_y^2 &= \frac{1}{1} \left(\sum_{n=1}^2 \overline{y}_n^2 - \frac{1}{2} \left[\sum_{n=1}^2 \overline{y}_n \right]^2 \right) \\ \sigma_y^2 &= \sum_{n=1}^2 \overline{y}_n^2 - \frac{1}{2} \left[\sum_{n=1}^2 \overline{y}_n \right]^2 \\ &= \overline{y}_2^2 + \overline{y}_1^2 - \frac{1}{2} \left[\overline{y}_2 + \overline{y}_1 \right]^2 \\ &= \overline{y}_2^2 + \overline{y}_1^2 - \frac{1}{2} \left(\overline{y}_2^2 + 2\overline{y}_2\overline{y}_1 + \overline{y}_1^2 \right) \\ &= \frac{1}{2} \left(\overline{y}_2^2 - 2\overline{y}_2\overline{y}_1 + \overline{y}_1^2 \right) \end{aligned}$$

$$= \frac{1}{2}(\overline{y_2} - \overline{y_1})^2$$

In general, for the sets of averaged samples $\overline{y_n}$ and $\overline{y_{n+1}}$ the general equation for 2 sample variance is given by:

$$\sigma_y^2 = \frac{1}{2}(\overline{y_{n+1}} - \overline{y_n})^2$$

This expression is fundamental to the definition of Allan Variance.

Allan Deviation and Variance.

D. W. Allan worked for many years with a group, led by J A Barnes, at the National Bureau of Standards in Boulder, Colorado, USA. In the 1960's they were experimenting with and developing atomic clocks. The group encountered problems when attempting to characterise the phase and frequency noise of oscillators associated with the atomic clocks but in 1965 David Allan showed that application of (a particular method of) two sample variance afforded a solution to their problems. The particular method used was later named after him.

Their work naturally extended to the general study of high frequency clock and oscillator stability. Experimental evidence demonstrated a logarithmically dependent divergence in the *Standard Deviation* of frequency fluctuations of high frequency oscillators which was attributed to some components of phase noise. Regardless of the number of sample measurements in large sets of phase (or frequency) measurements made on such clocks or oscillators, the calculated average values of repeated experiments refused to converge to a well defined mean value.

Equivalently, regardless of the length of time over which the parameter of interest was integrated the calculated averages (of repeated experiments) never converged to a well defined mean value.

As a result, characterisation of the noise for these clocks and oscillators with sufficient certainty was difficult to achieve.

A solution was afforded by the two sample variance:

$$\sigma_n^2 = \frac{1}{2}(\overline{y_{n+1}} - \overline{y_n})^2$$

The generalised two sample-variance gives a figure for the deviation of an averaged measurement of y_{n+1} from a previous average or estimate y_n . For the purposes of the calculation $\overline{y_n}$ represents the expectation of future y and in particular y_{n+1} . Indeed, it represents the expectation of the next *averaged* value $\overline{y_{n+1}}$, which is unlikely to be equal to $\overline{y_n}$ since the associated noise is not convergent!

For N sample averages a set of variances can be constructed by calculating (N-1) successive two sample variances:

$$\sigma_{y_n}^2 = \frac{1}{2}(\overline{y_{n+1}} - \overline{y_n})^2 \quad \forall n \in [1, N - 1]$$

where each two sample-variances is calculated using the previously measured mean value. (Hence (N – 1) iterations, each of which can be regarded as an individual experiment). From this set of variances an expectation value can be calculated

$$\begin{aligned} \sigma_y^2 &= \langle \sigma_{y_n}^2 \rangle = \frac{1}{2} \langle (\overline{y_{n+1}} - \overline{y_n})^2 \rangle \quad \text{or} \quad \frac{1}{2} E[(\overline{y_{n+1}} - \overline{y_n})^2] \\ &= \frac{1}{2(N-1)} \sum_{n=1}^{N-1} [\overline{y_{n+1}} - \overline{y_n}]^2 \end{aligned}$$

This is the fabled Allan Deviation which has the advantage over the Standard Deviation of being convergent for most types of clock noise. It is simply the variance within a set of ‘two sample averaged variances’ for some constant (sample or time) averaged period. Indeed, a range of variances can be calculated for a range of sample or time averaged periods.

By plotting the graph of Allan Deviation against progressively larger (sample or time) averaged periods it is possible to characterise the observed noise and identify types of noise present in the recorded signal, determine the band of frequencies over which these noise components are observable, quantify the noise and determine optimum measurement sample averaging rates and times.

A typical Allan deviation graph will give an indication of the optimum level of noise correction achievable by averaging, assuming ideal conditions, for a given system.

Allan deviation was first used predominantly to characterise frequency and phase noise *in the time domain* for atomic clocks and oscillators. As more applications were found, in frequency and phase noise characterisation of high precision oscillators and clocks and in particular for examining phase noise in GHz applications, a number of variants of the algorithm were developed (the most notable being the Modified Allan Variance^{*}). Allan Deviation has since been applied extensively in other fields of applied sciences to quantify noise present on the output of measuring systems that operate in the frequency or time domain, to characterise time dependent noise in sensors and amplifiers and to optimise noise levels in discretely sampled signals on systems of digital to analogue and analogue to digital conversion.

^{*}Note: The Modified Allan Variance was developed to overcome the inability of the standard Allan Deviation, when used to accurately determine clock and oscillator frequencies, to separate white phase modulation from flicker or 1/f phase modulation. This subject is left to the reader as *additional reading*.

Application to Real World Discrete Time Systems.

In general, a real world system will have a dynamic input. Averaged measurements on its output have to be made within some realistic period of time, so there must be some finite number of samples in each calculated average. The aim of characterising the noise on the output of a system (using Allan Deviation) is to determine the optimum number of samples in each calculated average that will provide the best possible attenuation of output noise.

To do this we need to characterise how the noise changes as a function of time over which samples are averaged (or over which continuous measurements are integrated) and by how much successive averages deviate from one another for a given averaging time. With that information the optimum sampling rate and averaging duration that will provide optimum noise correction can be determined.

Consider any real world electronic sensor that measures some parameter (distance, velocity, acceleration, density, whatever...) and which provides a calibrated output voltage, over its specified range, whose value is some well behaved function of the input parameter; i.e. the output is in some way proportional to the input for all input values of interest.

Since it is a real world sensor we can be absolutely sure that the output voltage will be a superposition of both the sought after signal and noise. In general the noise will be from a number of sources and its effect will be to obscure the true value of the parameter being measured.

From statistical analysis we can be reasonably sure, though by no means certain, that an average of two measurements will reduce the noise on the output by some modest amount. We also know that the larger the number of sampled measurements, the smaller we can expect the noise component to be in the recovered averaged output signal.

Provided the noise is convergent, then with a large number of samples over a long enough time the measure of noise will tend to zero and the output of the averager will tend toward the true value of the signal.

Even if the noise is not convergent, there will be components of the noise that are and these components will, over a long enough time, tend to average to zero, leaving only the signal superimposed with noise from non-convergent sources.

In general, for a test system with its input held constant and with observable noise with peak amplitude N_0 :

- a) Every individual measurement will be uncertain by $\pm N_0$ volts.
- b) The average of every two samples will be uncertain by $\pm N_0/2$ volts.
- c) The average of s samples will be uncertain by $\pm N_0/s$ volts.

Where the test system input is not held constant, time average measurements of its output will vary in time and will be indistinguishable from noise. This behaviour will adversely affect the test results and hence any subsequent output signal we intend to rid of noise.

Since the test system is being used to determine optimal noise related parameters it is reasonable to set the specific test condition that the input signal be held at some constant value, preferably zero, over the sampling time. The long time averaged sensor output can then, in general, be expected to approach its corresponding constant (or zero) output value. However this may not be the case as systematic sources of error may be present causing some small output offset (which may be time dependent over times greater than our averaging period) or there may be some non-convergent components of the noise present. Nevertheless, we can conclude that noise can be minimised and that the resultant measured and averaged output will be the best possible noise reduced output when the sensor is placed in a real world operating environment free of any stimulus. In practice the long time average may not be attainable but it will be the best possible and therefore represents an 'expectation value' to be aimed for.

Noise Characterisation.

There are two fundamental pieces of information needed to characterise the noise response of a sensor or system by Allan Deviation.

Both pieces of information are dynamic changes in the measured noise (which are normally but not necessarily functions of time) in the sensor/systems real world operating environment. Note that it is a necessary condition that dynamic characterization of the sensor/system and its noise be performed under the same conditions in which the sensor/system will be used. After all, our aim is to determine the characteristics of the noise on the output of the sensor/system, some sources of which may be environmental. It must be understood that we are not characterising the noise present in any particular sampled data set.

Remember: For noise characterisation the input parameter that the sensor or system is intended to measure must be held constant and preferably at zero.

Under these conditions the first piece of time related information we need is the Allan Variance (which is a function of the set of all two sample variances) in the system's output as a function of a discrete time average with (s) representing the number of samples used to calculate the sample average (or the number of units of time over which an average is calculated by integration) of each $\overline{y_n}$:

$$\sigma_n^2(s) = \frac{1}{2} \left[\overline{y_{n+1}(s)} - \overline{y_n(s)} \right]^2 = \text{two sample variance}$$

$$\sigma^2(s) = \frac{1}{2(N-1)} \sum_{n=1}^{N-1} \left[\overline{y_{n+1}(s)} - \overline{y_n(s)} \right]^2 = \text{Allan Variance}$$

For the first case, where no averaging is performed (or where averaging by integration is performed over the shortest integrating interval), the Allan variance is the standard deviation squared for all 2 sample variances between successive discrete samples.

$$\sigma_n^2(1) = \frac{1}{2} \left[y_{n+1}(1) - y_n(1) \right]^2 = \text{two sample variance}$$

$$\sigma^2(1) = \frac{1}{2(N-1)} \sum_{n=1}^{N-1} \left[y_{n+1}(1) - y_n(1) \right]^2 = \text{Allan Variance}$$

For the second case, where averaging is performed on consecutive pairs of samples (or where averaging by integration is performed over two units of integration time):

$$\sigma^2(2) = \frac{1}{2(N-1)} \sum_{n=0}^{N-1} \left(\overline{y_{n+1}(2)} - \overline{y_n(2)} \right)^2$$

where N is the number of averaged samples ($\overline{y_n}$) constructed from the set of M measured samples (x_m).

[Note that N is dependent on the method used to construct the averaged samples. If there are $M = 6$ samples of x_m and each $\overline{y_n}$ is an average of 2 samples then N can be:

- a) 3, for averages $y(x_1, x_2)$, $y(x_3, x_4)$ and $y(x_5, x_6)$. Therefore $N = M/2$. (Referred to as a non-overlapping average)
- b) 5, for averages $y(x_1, x_2)$, $y(x_2, x_3)$, $y(x_3, x_4)$, $y(x_4, x_5)$ and $y(x_5, x_6)$. Therefore $N = M - 1$. (Referred to as an overlapping average)

This implies that for choice a) there are $N = M/2$ sizes of average ranging from 1 sample per average to 3 samples per average and for choice b) there are $N = M - 1$ sizes of average ranging from 1 sample per average to 5 samples per average.

These represent two different methods of calculating Allan deviation using non-overlapping averaging and overlapping averaging.

In general, if each average is calculated from s samples then there are

- a) $N = M/s$ non-overlapping sample averages.
- b) $N = M + 1 - s$ overlapping sample averages.

The preference in this presentation is to use the overlapping option, with $N_{max} = M - 1$.

The paper Analysis of Geodetic Time Series Using Allan Variances: Thomas Friederich, Universität Stuttgart Geodätisches Institut gives an excellent graphical example of the difference in resolution between non-overlapping and overlapping sample averaging when performing Allan Deviation on a set of geodetic data.]

For successive cases: Each successive Allan Deviation is calculated in a similar manner. From the set of M samples calculate the set of averages that use s samples in each average. There will be a total of S sets of averages, each set being indexed s , where $s \in [1, M - 1]$.

Each set will contain N averages, where $N = 2$ for $s = M - 1$ and $N = M$ for $s = 1$. Therefore, the s 'th set of averages will contain $N = M + 1 - s$ overlapping sample averages, each indexed n , where $n \in [1, N]$.

Therefore, where the $y_n(s)$ and $y_{n+1}(s)$ are the n and $n + 1$ averaged values of s 'th set of averages and $s \in [1, S]$ represents the number of elements over which each average in the $S = M - 1$ sets of averages is calculated, then

$$\sigma^2(s) = \frac{1}{2(N-1)} \sum_{n=1}^{N-1} \left(\overline{y_{n+1}(s)} - \overline{y_n(s)} \right)^2$$

(Note: The total number of averages N in any set of averages is inversely proportional to the number of samples s used to calculate the individual averages in the set).

With this information we can determine the second (and very significant) piece of time related information which is the variance of the S Allan variances in the system's time averaged output. This is performed by a graphical analysis rather than by calculation.

The S Allan Variances are plotted against the number of samples s used to calculate the average in each 2 sample variance of the s 'th Allan Variance (or plotted against the corresponding sampling averaging time taken to acquire s samples).

For case 1, which can be expected to have the highest deviation, $\sigma^2(1)$ is plotted against 1 or the corresponding time taken to acquire each of the N samples, since 'averaging' of each $\overline{y_n}$ is done over 1 sample or 1 unit of time.

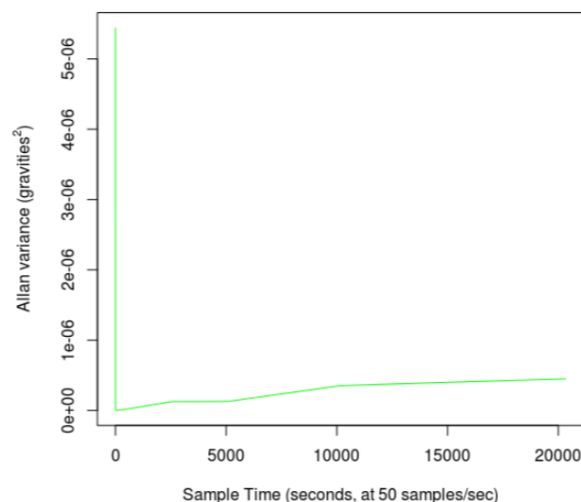
For case 2, which can be expected to have a slightly lower deviation by virtue of the averaging process, $\sigma^2(2)$ is plotted against 2 or the time taken to acquire each pair of samples, since averaging of each $\overline{y_n}$ is done over 2 samples or 2 units of time.

For successive cases, the Allan deviations $\sigma^2(s)$ may be expected to have progressively lower deviations as s increases, by virtue of the larger number of samples in each averaging process. Each $\sigma^2(s)$ is plotted against s or against the time taken to acquire each set of samples, since averaging for each $\overline{y_n}$ is done over a set of s samples or s units of time.

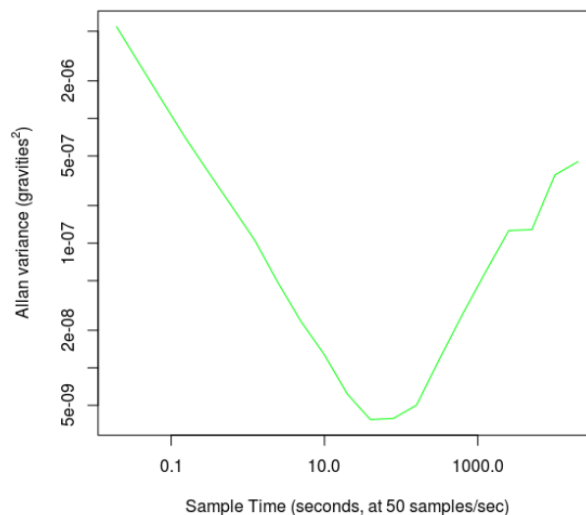
One would expect the deviations to initially fall off very quickly and eventually decay exponentially to zero. In practice though, in many applications with a large enough number of samples, the graph does indeed fall off very quickly but it can often decay to some clearly observed minimum and then, perhaps surprisingly, begin to rise again very slowly as the deviation averaging period rises.

The significance of the minimum is that it lies at the optimum sample averaging time (or number of samples per second) for any noise reduction process used with the particular sensor/system.

The graph below is an example of an Allan Variance plot from an experiment that recorded many sample measurements of gravity over a period of 20000 seconds using an accelerometer that was susceptible to many sources of vibrational, acoustic and thermal noise.



The very sharp initial fall and subsequent slow rise makes it difficult to see definition in the plot. However, by plotting the logarithm of the variance on the y axis (noting that $\text{Log}\sigma^2(s) = 2\text{Log}\sigma(s)$) and plotting the sampling time on a logarithmic x axis (because deviation falls off very quickly over the first few points in the data set) a log-log plot of the Allan Variance gives much better definition to the curve.



Interpretation of the Allan Variance Curve.

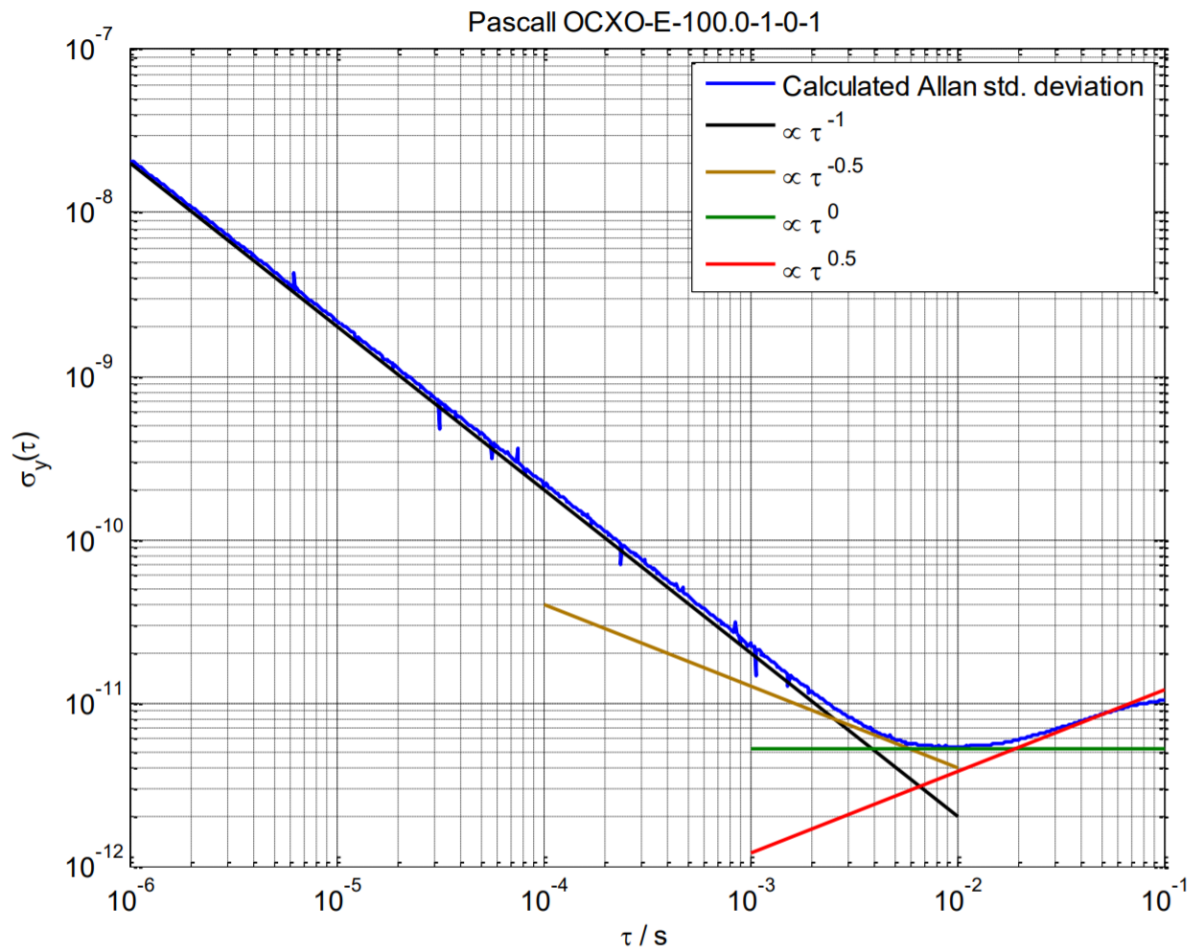
There are essentially three distinct parts to the Allan Variance curve.

The first is that part of the graph with the negative gradient slope falling from the first point, $\sigma^2(1)$ to the points that are in close proximity to the minimum. Each point represents the deviation or variance of a set of averaged values constructed from the data set; for instance *the second point* on the Allan Variance curve would be the set of running averaged values $(x_m + x_{m+1})/2 \quad \forall i \in \mathbb{N}$.

The effect of averaging over 2 samples suppresses the highest frequency noise components so the deviation represented by the second point on the curve is (in general) less than the first, the third less than the second and so on.

The second is the area in the neighborhood of and including the minimum. One would expect that the effect of averaging over larger numbers of samples would have an increasingly beneficial effect on the variance (and deviation) of the set of averaged samples and that it would exponentially tend to zero. However for systems with non-convergent noise it doesn't, for a number of reasons.

The third is that part of the graph with the positive gradient slope rising from the minimum. The main reason for the rise is that at low frequencies and long sampling times $1/f$ noise (as well as other low frequency noise sources such as random walk) become significant. Sample averages of very low frequencies, averaged over less than a full cycle will, in general, have a distribution of values from the maximum amplitude of the low frequency components to the minimum. Over a band of low frequencies this naturally leads to a measurable noise distribution.



Graph of Allan Deviation for an oven controlled oscillator with marked up noise processes.

We have a total measurement time T_{meas} over which n samples have been measured and/or calculated.

We have a sampling time T_s which is a constant for any particular Allan Deviation.

For a particular sampling time T_s there will be a maximum number of time averaged samples n_s that can be made during the measurement period T_{meas} .

Therefore $T_{\text{meas}} = n_s T_s$.

As T_s gets longer the HF cut off frequency of the algorithmic low pass filter gets lower.

For T_1 (shortest sample averaging time) with n_1 being the max num of samples in T_{meas} , the noise contributing to the Allan deviation is the sum of all noise frequency components in the signal whose highest frequency is limited by the sample averaging time.

For $s > 1$ the highest frequencies are averaged away (for convergent noise!) and the variance due to noise becomes more and more the sum of the noise from only lower and lower frequency components.

The two sample deviation measures deviation between successive averaged samples. So deviation due to noise at frequencies equal to or greater than $1/(\text{two sample periods} + \text{any dead time})$ will be fully captured. Deviation due to noise at frequencies somewhat lower than $1/(\text{two sample periods} + \text{any dead time})$ will only capture that due to rises or falls in the signal during that period (which is less than the period of the noise frequency component of interest). The lower the noise frequencies of interest for a given sampling frequency the smaller will be the two sample deviation; the lower the sampling frequency for a given noise frequency of interest the greater the deviation will be (provided the sampling frequency is greater than the noise frequency of interest OR provided the sampling period is less than the period of the noise frequencies of interest.)

Since amplitude (or power) of $1/f$ noise rises inversely with frequency then the two sample deviation will rise as the sampling period increases (or as the sampling frequency falls).

Note: There is a low frequency cut off set by T_{meas} . For short T_{meas} , frequencies significantly lower than $1/(T_{\text{meas}} \times 4)$ [$1/4$ wavelength] will contribute little toward any signal deviation as the noise at these low frequencies is very slow changing.

Only when we start looking at signals on long timescales does the variance of low frequency noise become significant. And since the contribution to the total noise is predominantly from $1/f$ noise sources at low frequencies then in systems that exhibit $1/f$ noise the variance will grow as T_{meas} grows.

For many systems low frequency responses are of no interest (e.g. audio and RF amplifiers and sensors) with LF cut off being built into the design. But with very slow changing time dependent systems $1/f$ noise and related sources of non-convergent noise become significant. For example, the slow changes in frequency of an oscillator in response to variations in temperature, which may fluctuate over 24 hour periods (daily) and/or over the period of a year (seasonal). Another example is a gravimeter which is subject to low frequency noise from the relative motions of celestial objects.

Oscillators and frequency stability.

Phase Noise

A perfect oscillator would have a single stable frequency and its frequency spectrum would manifest as a delta-function. However real world oscillators exhibit random phase variations, generally referred to as *phase jitter*. The phase jitter gives rise to a spread of observable frequency components in the oscillator's output which appear as sidebands in its frequency spectrum.

The term *phase noise* is used to describe the spectral noise that results from phase jitter.

Since phase and frequency are intimately related, phase noise can also be regarded in terms of random frequency variations.

Phase noise is concerned with a signal's frequency spectrum and is therefore considered to be in the frequency domain. In contrast, phase jitter is concerned with variations of phase as a function of time.

Phase noise and phase jitter specification are important parameters in many frequency dependent systems.

For example, modern radio transmitter and receiver systems predominantly employ phase angle modulation techniques and the presence of phase noise on their local oscillators and mixer/heterodyne circuits can seriously affect their performance and hence the system's specifications. Similarly, atomic clocks and high speed data transmission systems require ultra-stable oscillators with exceptionally low values of jitter to minimise system errors.

Allan Deviation and Variance of Oscillators and Clocks

Allan deviation and variance was first used to characterise phase noise of oscillators where the measured frequency was observed to change in a random manner from moment to moment and measurement sample to measurement sample (for whatever reason. It is after all a type of noise!).

The frequency stability of an oscillator can be determined from the oscillator or clock model:

$$A(t) = [A_0 + \epsilon(t)]e^{i(\omega_0 t + \phi(t))}$$

Where

- $A(t)$ =Clock output amplitude,
- A_0 =Nominal peak amplitude,
- $\epsilon(t)$ =Amplitude deviation,
- ω_0 = Nominal angular frequency,
- $\phi(t)$ =Phase deviation or phase error.

Time domain frequency analysis of clocks and oscillators is primarily concerned with changes in frequency due to the phase deviation term $\phi(t)$. The amplitude deviation is unimportant, negligible and can be approximated to zero.

$$\begin{aligned} A(t) &= A_0 e^{i(\omega_0 t + \phi(t))} \\ \frac{dA(t)}{dt} &= \frac{d i(\omega_0 t + \phi(t))}{dt} \cdot A_0 e^{i(\omega_0 t + \phi(t))} \\ &= i(\omega_0 + \frac{d\phi(t)}{dt}) \cdot A_0 e^{i(\omega_0 t + \phi(t))} \\ &= i\omega \cdot A_0 e^{i(\omega_0 t + \phi(t))} \end{aligned}$$

Where

$\omega = \omega_0 + \frac{d\phi(t)}{dt}$ is the instantaneous angular frequency

$\nu(t) = \nu_0 + \frac{1}{2\pi} \frac{d\phi(t)}{dt}$ is the instantaneous frequency.

And the (instantaneous) frequency difference due to noise is given by

$$\Delta\nu = \nu(t) - \nu_0 = \frac{1}{2\pi} \frac{d\phi(t)}{dt} \quad (\text{Eq.1})$$

The fractional frequency is a basic measure of the accuracy of a precision clock or oscillator which is defined as the ratio of the change in frequency (due to noise) and the rated or expected frequency:

$$y(t) = \frac{\Delta\nu}{\nu_0} = \frac{\nu(t) - \nu_0}{\nu_0}$$

Substituting with the rate of change of phase term in Eq1 gives

$$y(t) = \frac{1}{2\pi\nu_0} \frac{d\phi(t)}{dt} = \frac{d}{dt} \left[\frac{\phi(t)}{2\pi\nu_0} \right] = \frac{d}{dt} \left[\frac{\phi(t)}{\omega_0} \right] = \frac{dx(t)}{dt} \quad (\text{Eq.2})$$

Where the term $x(t) = \frac{\phi(t)}{\omega_0}$ is commonly referred to as the random time deviation, the phase time error, jitter or time fluctuation.

Time domain clock or oscillator stability analysis relies on the acquisition of an array of M temporally equal spaced phase measurements, x_m or equivalently an array of $M - 1$ fractional frequency deviations, y_m . (Note though that the x_m are functions of time whilst the y_m are dimensionless.)

The measurement interval between samples is the acquisition system's sampling time τ_0 .

Where analysis is performed on average values of y calculated over S samples with an averaging time or observation interval $\tau = S\tau_0$ ($S \geq 1: S \in \mathbb{N}$), the average fractional frequency can be given by:

$$\overline{y_m(\tau)} = \frac{1}{\tau} \int_{t_m}^{t_m + \tau} y(t) dt \quad \text{for a continuous function } y(t)$$

Or

$$\overline{y_m(S\tau_0)} = \frac{1}{S} \sum_{m=i}^{i+S} x_m$$

for a discrete set of m samples over the observation interval $\tau = S\tau_0$.

Note: When using consecutive overlapping sets of $\{x_m\}$, each containing S members x_m , there can only be a total of $N = M - \frac{\tau}{\tau_0} = M - S$ values of $\overline{y_i}$

Where analysis is performed over successive sample pairs the y_m correspond to the first differential of $x(t)$ WRT time:

$$y_m = \frac{x_{m+1} - x_m}{t_{m+1} - t_m} = \frac{x_{m+1} - x_m}{T} \quad \text{where } T = \tau_0 + \text{dead time.} \quad (\text{Eq.3})$$

And the second order difference of $x(t)$ WRT time is given by:

$$z_m = \frac{y_{m+1} - y_m}{t_{m+1} - t_m} = \frac{x_{m+2} - 2x_m x_{m+1} + x_m}{T^2}$$

Similarly, where analysis is performed over successive pairs of sample averages the $\overline{y_m}$ correspond to the first differential of $x(t)$ WRT time:

$$\overline{y_m} = \frac{\overline{x_{m+1}} - \overline{x_m}}{t_{m+1} - t_m} = \frac{\overline{x_{m+1}} - \overline{x_m}}{T} \quad \text{where } T = \tau_0 + \text{dead time.} \quad (\text{Eq.4})$$

And the second order difference of $x(t)$ WRT time is given by:

$$z_m = \frac{\overline{y_{m+1}} - \overline{y_m}}{t_{m+1} - t_m} = \frac{\overline{x_{m+2}} - 2\overline{x_m} \overline{x_{m+1}} + \overline{x_m}}{T^2}$$

Note: The variance depends on the sampling time over which the sample is measured (or calculated) in comparison to the long time rate of change of the noise phenomena.

In fact the deviation algorithm is a low pass filter whose cut-off frequency is set by the number of samples. (Other variances, such as the Modified Allan Variance, have lower cut off frequencies that smooth away longer term fluctuations and can demonstrate a variance that is a function of measurement time, τ .)

Calculation of the Allan Deviation for a Clock or Oscillator

Where τ_0 is the sampling period, a set of M phase measurements can be made in discrete time with one measurement being made every τ_0 seconds, therefore

$$\phi_i(t) = \phi(m\tau_0) \quad m \in [1, M]$$

From which a set of $M - 1$ fractional frequency estimates can be made, assuming there is no dead time, using Eq.2 and Eq.3

$$y_m = \frac{x_{m+1} - x_m}{t_{m+1} - t_m} = \frac{x_{m+1} - x_m}{\tau_0}$$

If we set a measurement averaging period $\tau = S \cdot \tau_0$, a set of $N = M - S$ overlapping averages, $\overline{y}_n$, can be calculated from the set of y_m fractional frequency estimates

$$\overline{y}_n = \frac{1}{S} \sum_{s=n}^S y_s = \frac{1}{M-N} \sum_{m=1}^{M-N} y_{n+m}$$

The Allan Deviation of the oscillator or clock frequency can then be calculated from the set of N averages, $\{\overline{y}_n\}$:

$$\sigma^2(\tau) = \frac{1}{2(N-1)} \sum_{n=1}^{N-1} \left(\overline{y_{n+1}}(\tau) - \overline{y_n}(\tau) \right)^2$$

References:

https://cdn.rohde-schwarz.com/pws/dl_downloads/dl_application/application_notes/1ef69/1EF69_4e_TD_Osc_Stability_Allan.pdf

gives a good description of its application and how it should be applied depending on the time interval between successive frequency measurements. Where τ is long use average of frequency over τ . Where t is short use successive delta phase measurements.

<https://elib.uni-stuttgart.de/bitstream/11682/3866/1/Friederichs.pdf>. Analysis of Geodetic Time Series Using Allan Variances; Thomas Friederichs; Universität Stuttgart, Geodätisches Institut.

https://www.phidgets.com/docs/Allan_Deviation_Primer#Allan_Variance Allan Deviation Primer.

Statistics of Atomic Frequency Standards; David W. Allan; Proceedings of the IEEE, Vol 54, Feb 1966.

Atomic Timekeeping and the Statistics of Precision Signal Generators; James A. Barnes; Proceedings of the IEEE, Vol 54, No 2, Feb 1966.

Characterization of Frequency Stability *

James A. Barnes, Andrew R. Chi, Leonard S. Cutler, Daniel J. Healey, David B. Leeson, Thomas E. McGunigal, James A. Mullen, Jr., Warren L. Smith, Richard L. Sydnor, Robert F. C. Vessot and Gernot M. R. Winkler; IEEE Transactions on Instrumentation and Measurement; May 1971.

On the measurement of frequency and of its sample variance with high-resolution counters; Enrico Rubiola; Université Henri Poincaré; Review of Scientific Instruments 76, 054703 s2005d.

The Science of Timekeeping; Hewlett Packard; Application Note 1289

Addendum

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Dr Mark Hosey, 4th March 2020

This document was written by Dr Mark Hosey, started in February of 2019 and completed in May of 2019. Some small changes, corrections and additions have been made since then. The contents are the result of research stimulated after over hearing a discussion of a customer's use of Allan Deviation to illustrate a problem and the interpretations made about Allan Deviation by those discussing the problem. The work is free from plagiarism but is my interpretation and insight gained by reading material found published and freely available (if difficult to find) on the internet. My hope is that readers will share in my insights and find a simplicity in the principles that is not conveyed by the aforementioned freely available information.